AI for Healthcare

Task-2

Week-2

Dataset 1 : Chest X-ray14(NIH)

1) Type of Imaging Data:

Chest X-rays (Frontal view, PA/AP projections).

2)Number of Images:

112,120 X-ray images from 30,805 unique patients.

3)Available Classes or Labels:

14 distinct lung pathology labels (e.g., Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumonia, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, and Hernia), along with a "No Finding" class. It is a multi-label dataset (An image can have more than one label).

4)Dataset Imbalance:

Highly imbalanced. The 'No Finding' class accounts for over 50% of the data (60,000 images). Among the pathologies, 'Infiltration' and 'Effusion' have over 10,000 cases each, whereas rare conditions like 'Hernia' have fewer than 300 instances.

5)Observed Challenges:

Label Noise: The labels were automatically extracted from text radiology reports using Natural Language Processing (NLP). This introduces an estimated 10% error rate, meaning some annotations are inaccurate.
Bounding Box Scarcity: Only a tiny fraction (less than 1,000 images) contains hand-annotated bounding boxes for localization; the rest are strictly image-level labels.

Dataset 2: HAM10000 (Kaggle / Harvard Dataverse)

1)Type of Imaging Data:

Dermatoscopic (skin lesion) images.

2)Number of Images:

10,015 dermatoscopic images.

3)Available Classes or Labels:

7 categories of pigmented skin lesions:

1. Actinic keratoses and intraepithelial carcinoma (akiec)
2. Basal cell carcinoma (bcc)
3. Benign keratosis-like lesions (bkl)
4. Dermatofibroma (df)
5. Melanoma (mel)
6. Melanocytic nevi (nv)
7. Vascular lesions (vasc)

4)Dataset Imbalance:

Extremely skewed. Melanocytic nevi (common moles) dominate the dataset with 6,705 images (over 65% of the total data). In contrast, dermatofibroma (df) has only 115 images, making it difficult for models to learn the minority classes effectively without heavy data augmentation.

5)Observed Challenges:

Visual Artifacts: The images frequently include distracting artifacts such as skin hair, air bubbles from the immersion fluid, marker ink, and varying skin tones/lighting conditions.
Verification Biases: More than half of the dataset was validated purely by follow-up or expert consensus rather than the gold standard of histopathology, which can leave room for diagnostic ambiguity.